Week 7 Project Summary

Data Science track · HealthConnect Rwanda · Prepared by Emmanuella Esi Arhin

#	Item	Summary
1	Planned to test	The Week 6 candidate classifier (admission proxy), its preprocessing and its interface, against criteria fixed in advance.
2	Actually tested	12 tests (reproducibility, locked hold-out, 10 repeated CV runs, 100-shuffle permutation test, over-fitting, time split, calibration, operating point, errors, 50 segments, feature reliability and leakage, target scan with positive controls) plus 15 interface contract tests and 5 analytical-finding validations.
3	Most important results	Reproduces exactly (AUC 0.505) but hold-out AUC 0.479 (CI 0.423-0.535), permutation p = 0.42, time-split AUC 0.467. Positive controls prove the harness detects signal, so the null result is a data property.
4	Issues / weaknesses	Over-fitting (train 0.75 vs validation 0.51); unreliable features; F1 threshold flagging 95%; no release gate; silent interface failures (10/15 contract tests); uneven flagging by age band; no signal in the data.
5	Improvements made	Reliable-feature set; regularised model chosen on training data only; top-20% flag-rate budget; executable release gate; validated scoring wrapper with schema and per-row status.
6	Retest results	Over-fitting gap 0.24 to 0.06; contract tests 10/15 to 15/15; hold-out AUC 0.455 (CI 0.398-0.513); permutation p 0.73; gate still FAIL.
7	Tracks collaborated with	Data Analytics (Olabisi Amusan): two-way validation. Project Management: issue log and readiness. ML Engineering: interface test done on the Data Science side only; no counterpart available.
8	Tested collaboratively	With Data Analytics: Olabisi independently confirmed the seven overlapping Week 6 findings (coverage, facility conflict, no attendance field, weak signal, balanced classes, rare repeat visits, unusable fields), added a staff-record facility conflict, and ran three checks I requested against the source files: my 88% Follow-Up finding (confirmed, 88.0%), DoctorYears (clean join but department/facility link about random; kept as a doctor-level attribute only) and the target and HasChronic coding (confirmed; target rate 10.88% vs 10.71% expected if its two parts were independent).
9	What changed as a result	Facility, department, BMI and department-workload inputs removed; model now scored only through a validating wrapper; new follow-up/prior-visit finding raised with Data Analytics.
10	Key findings	No feature, segment or alternative outcome carries usable signal; the admission proxy cannot be predicted from these tables; the data behave like simulated data.
11	Challenges	Only 108 positives in the locked test set; no admissions or attendance table; Week 5 baseline not available; needing to distinguish a weak test from a genuine null (solved with positive controls).
12	Decisions	D1: Acceptance criteria fixed before testing; test set locked and used once per model version; D2: Model chosen on training data only; simplest model within 1 SE of the best (C. Ridge logistic, 10 reliable features); D3: Remove FacilityID, DepartmentName, BMI_clean, DeptWorkloadIndex; D4: Operating point = flag-rate budget (top 20%), reported with lift; D5: Do NOT release the model for any operational use; release only the tested interface and harness.
13	Remaining limitations	Small sample, proxy target, wide intervals, diagnosis timing, and no ML Engineering counterpart to run the interface tests.
14	Remaining issues / dependencies	No signal in the data (I6); no Admissions table (I7); no attendance/no-show field (I8); only 5,000 of 60,000 transactions link to an appointment (I9); prediction time-point undecided (I10); uneven flagging by age band (I12); no ML Engineering counterpart to run the contract tests (I5).
15	Contribution to HealthConnect	A reusable, proven test harness and release gate; a safe model interface for ML Engineering; a validated, evidence-based statement of what the data can and cannot support.
16	Before Week 8	Obtain outcome tables (admissions, referrals, attendance); ML Engineering runs the contract tests in its pipeline; agree the prediction time-point; Data Analytics repairs appointment keys; present the work as a validated null result plus reusable harness.